Automated CFD Workflow for a 2D Axial Turbine Stage

Transient rotor-stator interaction, systematic meshes and numerical verification

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Abstract

This study presents a reproducible ANSYS Fluent/PyFluent workflow for a planar axial turbine stage with a translating rotor and periodic sliding interface. The immutable historical solution, its baseline mesh, and a separate family of three systematic meshes are kept distinct. The workflow verifies native restarts, physical time, rotor position, conservation, actual inner iterations, mesh quality and sampled wall resolution. The systematic meshes contain 54,287, 104,831 and 196,569 active cells and use fresh Hybrid Initialization after restoring the original rotor position. They were calculated for 35, 40 and 35 blade-passing periods, respectively, using 24 MPI processes, 400 steps per period and 30 inner iterations. Statistics were recomputed directly from the accepted native monitor histories. M2-to-M3 mean differences are 0.028-0.138%, whereas peak-to-peak amplitudes differ by 5.86-7.52%. The means are weakly sensitive to the last refinement, but nonmonotonic three-mesh trends, remaining cycle-amplitude drift and unresolved timestep sensitivity prevent a general grid-independence or GCI claim. Source frames and a fine-mesh Mach movie retain fixed legends and the real physical clock. The simulated durations and remaining numerical limitations are stated explicitly. The study provides numerical verification checks; no experimental validation is claimed.

Keywords: axial turbine stage; rotor-stator interaction; sliding mesh; SST; Hybrid Initialization; PyFluent; mesh sensitivity; numerical verification

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1 Project objectives and study scope

The planar geometry comprises two fluid surfaces, one for the stator and one for the rotor, based on the public SU2 axial_stage_2D numerical reference. The original Fluent model provides the historical setup reference. Its existence, elapsed time and residual levels do not establish numerical independence or physical accuracy.

Three datasets remain separate: the immutable historical baseline with fixed topology; automated reproduction and iteration/timestep studies on that baseline mesh held fixed; and solutions on three newly generated systematic meshes. The initial 73167-cell mesh and the baseline mesh are not relabelled M1 and M2.

Saved-data integrity, periodic settling, timestep sensitivity and spatial sensitivity were assessed separately. A prescribed simulated duration was not treated as evidence of convergence. The three-mesh comparison was based on accepted native monitor histories and complete, phase-aligned statistical windows.

2 Workflow architecture and reproducibility

2.1 Software environment and run isolation

The workflow separates Python code from JSON configuration. Native solver setup is inherited from the verified case, rather than reconstructed from nominal SU2 values. The dedicated Python 3.12 environment uses ansys-fluent-core 0.42.2, ansys-api-fluent 0.7.0 and the Fluent 2024 R2 generated settings/TUI API. The tested dependencies are pinned in the environment lock file. Recorded hardware: Intel Core i9-14900K, 24 physical cores, 32 logical processors and approximately 64 GiB RAM on Windows 11 Pro. The systematic mesh campaign used 24 verified MPI ranks; no physical-core affinity optimisation is claimed.

Table 1: Workflow modules and responsibilities

Module / configuration	Responsibility
`workflow/common.py`; `workflow/cfd.py`	Portable paths, input integrity, isolated launch and offline entry points
`config/hybrid35_campaign_config.json`	Fresh original-position Hybrid campaign, timestep, iterations and per-run limits
Run progress and segment manifests	Accepted monitor segments, actual clock, checkpoints, stop reason and provenance
`workflow/assess_hybrid35.py`; verification tables	Cycle, field, conservation, iteration and sampled wall checks
`config/native_animation_config.json`; PNG indices	Fixed camera/legend, real clock, frame acquisition and hashes
`workflow/export_M3_verified_clip.py`	Offline encoding of the accepted 5T-35T interval
`docs/report/`	Final PDF documentation, prepared separately from the CFD workflow
`workflow/package_project.py`	Curated ZIP archive with a SHA256 manifest and verification

Health monitoring was used to detect non-finite fields, licensing errors, insufficient storage and numerical/runtime limits. Coherent case/data checkpoints and separate output directories were used throughout. Native computation was performed in double precision; field-data exports can contain float32 coordinates, which were accounted for in geometric comparison tolerances.

Run configurations, monitor segments and native checkpoint hashes were retained to provide traceability. Physical time, rotor position and the second-order time-history arrays were verified before continuation. Data belonging to different runs or solution times were not substituted for missing historical checkpoints.

2.2 Campaign sequence and restart policy

The workflow comprised baseline preservation and reopening, verification of setup and motion, an isolated smoke test, inner-iteration and timestep comparisons on the fixed baseline mesh, systematic mesh generation, restoration of the initial rotor geometry, Hybrid Initialization and separate integration of each mesh. Run directories and explicit segment boundaries were used to distinguish accepted monitor histories from discarded continuations.

3 Geometry, physical model and monitor definitions

3.1 Reference frame, pitch and rotor kinematics

Table 2: Verified geometry and reference quantities

Quantity	Verified value
Plane	XY; X axial, Y tangential; planar 2D
Pitch	0.04463756775 m
Rotor mesh velocity	Vy = -150 m/s; angular velocity = 0 rad/s
Blade-passing period	$T = p/\|V_y\| = 2.97583785 \times 10^{-4}$ s
Reference depth	1 m for reported Fluent forces and mass flow
Periodicity	Translational on both rows; nonconformal periodic sliding interface
Blade walls	Stationary relative to the adjacent mesh; no duplicated wall motion

Rotor position is checked using physical mesh coordinates. Fluent can wrap the entire rotor by an integer pitch; a position is therefore not inferred from -150*t alone. Comparisons use equal rotor phase, with integer-pitch translations verified geometrically. The clock phase t/T modulo 1 does not define a CAD zero position.

3.2 Physical and numerical settings

The verified settings are retained in a readable setup record and complete JSON export. SU2 reference values are distinguished from the settings confirmed in Fluent. Density, temperature, pressure and Mach fields are checked for finite values.

Native Mach was verified against velocity magnitude divided by sound speed (maximum discrepancy about 1.51e-7). It is the absolute Mach field, rather than relative Mach. Static pressure is absolute because operating pressure is zero. Native blade-wall Y velocity is zero on the stator and -150 m/s on the rotor, confirming that mesh and wall motion are not applied twice.

3.3 Monitor surfaces, weights, units and signs

A positive boundary mass flow denotes inward flux. Outlet flow is therefore negative. Tangential force is the global Y component on the rotor blade, at a reference depth of 1 m. Total pressure is

Table 3: Settings confirmed in the native Fluent case

Setting	Verified value
Solver	ANSYS Fluent 2024 R2; planar 2D; double precision; pressure based
Models	Unsteady RANS SST k-omega; energy and viscous dissipation enabled
Air	Ideal gas; Sutherland viscosity; cp 1006.43 J/(kg K); thermal conductivity 0.0242 W/(m K)
Operating pressure	0 Pa
Inlet	Total pressure 169623.33 Pa; total temperature 305.76 K; turbulence intensity 3%; viscosity ratio 100
Outlet	Static pressure 99741 Pa
Walls	No slip; adiabatic; stationary relative to the moving/stationary adjacent zone
Coupling and gradients	PISO; least-squares cell-based gradients
Spatial schemes	Second-order pressure; second-order upwind transported quantities
Time integration	Second-order implicit
Historical run	dt = 3e-6 s; nominal 30 inner iterations per step

Table 4: Native monitor definitions

Monitor	Definition and unit
Rotor tangential force	Y force on wall_rotore; N for 1 m depth; reported sign retained
Inlet mass flow	Inward-positive signed mass flow; kg/s
Outlet mass flow	Outward flow is negative in the inward-positive convention; kg/s
Outlet total pressure	Mass-flow-weighted total pressure at the outlet; Pa
Domain mass	Volume integral of density over both active fluid zones; kg
Wall data	Absolute static pressure and native y+ on blade-wall faces

mass-flow weighted at the outlet; operating pressure is zero, so native static pressure is absolute. A rotor total-pressure change includes work exchange and is not by itself a dissipative loss.

4 Mesh generation, topology and quality

4.1 Baseline mesh and fixed topology

All reported sensitivity calculations retain fixed mesh topology. Automatic mesh adaptation is disabled; rotor translation is handled by the sliding interface and does not imply mesh refinement.

The saved baseline contains 361839 stored cells: 289671 active fluid cells and 72168 inactive hierarchy parents. The active stator and rotor regions contain 64112 and 225559 cells, respectively. Inactive parents are excluded from solution-cell statistics. This baseline is separate from the M1/M2/M3 meshes generated through the controlled meshing procedure.

The stator fluid zone is fixed. The rotor fluid zone translates in Y. Their nonconformal sliding interface is sliding_statore_rotore; the periodic boundaries match within approximately 4.27e-9

m in the baseline geometry audit. Parent zones are recorded separately and are excluded from active-cell statistics. Reproduction keeps this topology and cell count fixed while the rotor mesh moves.

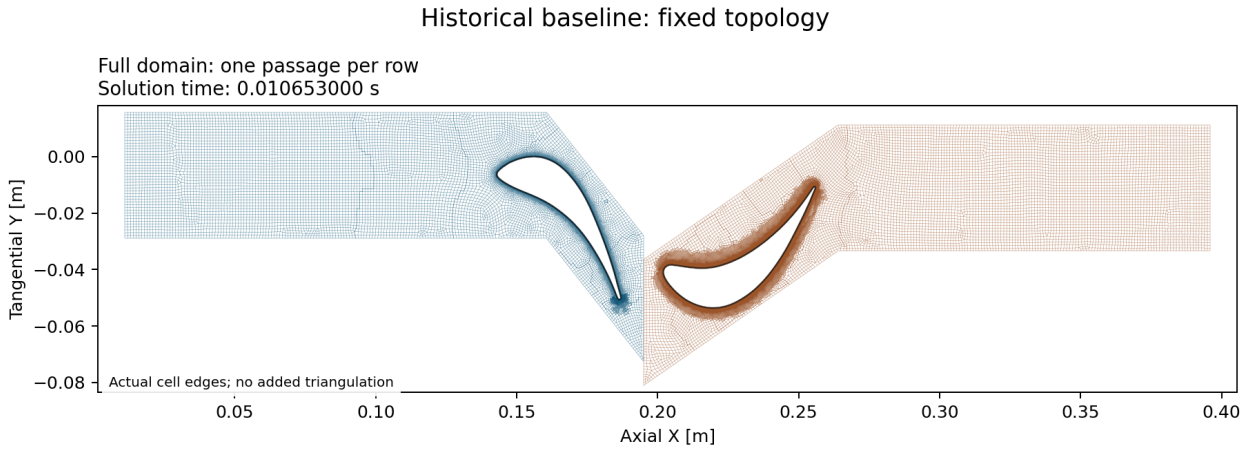


Figure 1: Historical baseline mesh at t=0.010653 s. Actual polygon edges are shown; inactive parent cells are excluded from active-cell statistics.

4.2 Systematic mesh family and wall layers

All three meshes use the same STEP and ANSYS Meshing procedure. Global sizing, curvature control, blade tangential spacing and first-layer height are refined together. The nominal inflation thickness remains 0.00094396216644 m. This is an anisotropic family; cell counts alone do not establish an asymptotic refinement sequence.

Table 5: Systematic sizing and inflation controls

Mesh	Global size [mm]	Blade spacing [mm]	First layer [micrometres]	Layers
M1	1.4	0.21	1	29
M2	1	0.15	0.7142857	41
M3	0.7142857	0.107142857	0.510204	57

Table 6: Native active-cell counts and mesh-quality extrema

Mesh	Active cells	Min orthogonal quality	Max aspect ratio	Max equiangle skewness
Historical baseline mesh	289671	0.0263166	76.9229	0.6994222
M1	54287	0.2461951	321.9638	0.7941735
M2	104831	0.4701662	214.5706	0.7216187
M3	196569	0.4183993	219.4238	0.7344144

The minimum orthogonal quality of the baseline mesh is only 0.0263166. The systematic meshes have positive cell volumes and minima of approximately 0.2462, 0.4702 and 0.4184. Quality is not monotonically improved by cell count. Their high aspect ratios occur in thin wall layers: all cells above aspect ratio 100 lie within approximately 11.36 micrometres of the corresponding blade, as checked against actual wall segments. This location does not replace a y+ or solution-quality check.

The direct systematic mesh exports are the authoritative meshes, stored locally as `inputs/meshes/M1_systematic.msh`, `M2_systematic.msh` and `M3_systematic.msh`. Historical Workbench FFF.msh caches contain 73167 cells and must not be substituted. Per-cell metrics, per-zone percentiles, threshold counts, extrema locations and native polygon-edge PNGs are retained with hashes in the technical archive. Selected quality tables and figures are supplied with the publication material. Raw-mesh screenshots and setup-verified screenshots are distinguished.

A bidirectional vertex-to-segment comparison checks actual blade-wall polylines at the baseline phase, without fitting a translation. The largest difference is 4.90 micrometres and includes curve tessellation differences. M2 discrepancies approach float32 coordinate resolution; these small values are not a claim of CAD accuracy at nanometre scale.

Table 7: Geometry agreement from bidirectional blade-polyline matching

Mesh	**Blade**	**Maximum polyline difference [micrometres]**
M1	Stator	4.897723
M1	Rotor	2.590972
M2	Stator	0.007679605
M2	Rotor	0.01489774
M3	Stator	2.627585
M3	Rotor	1.50755

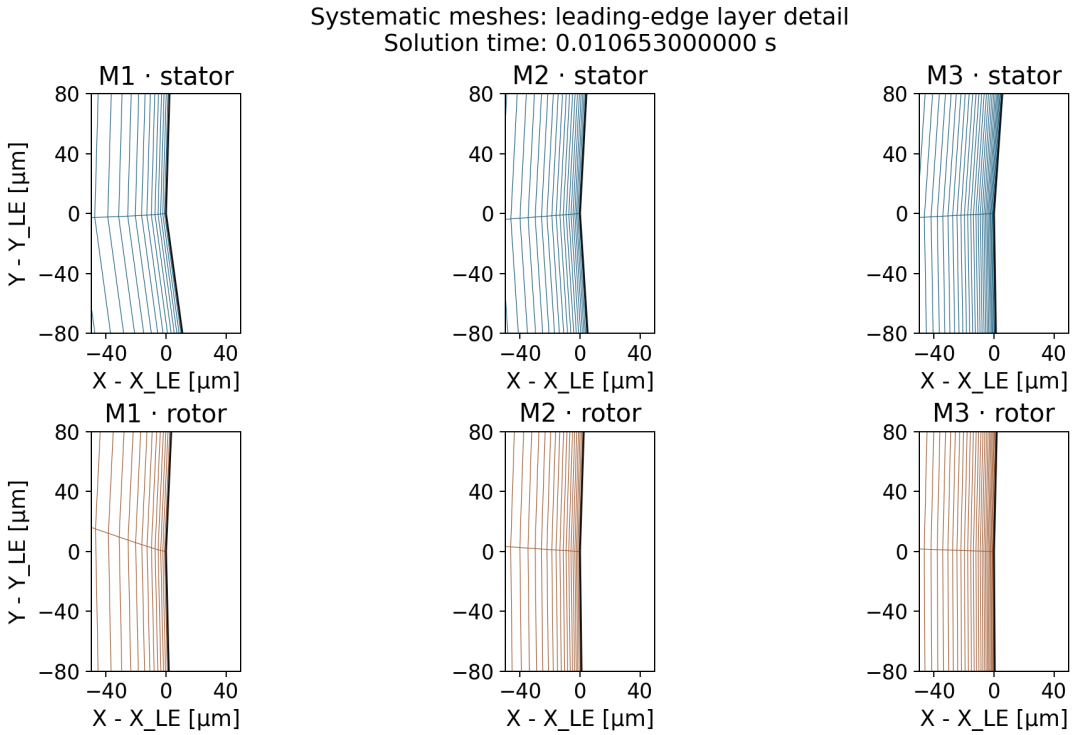
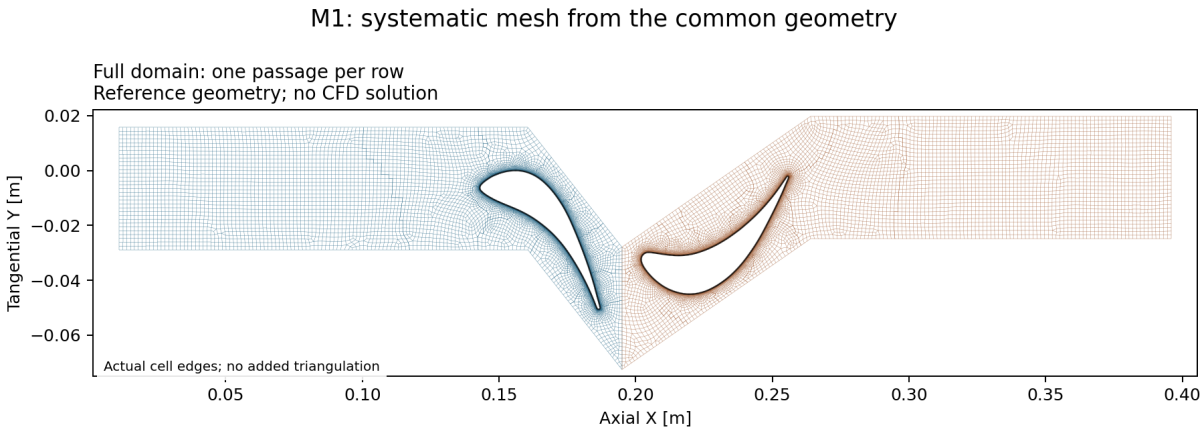
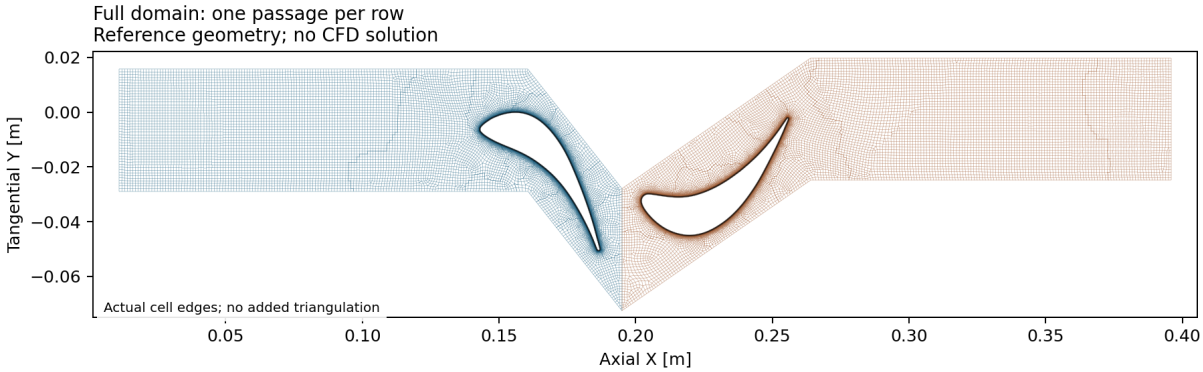


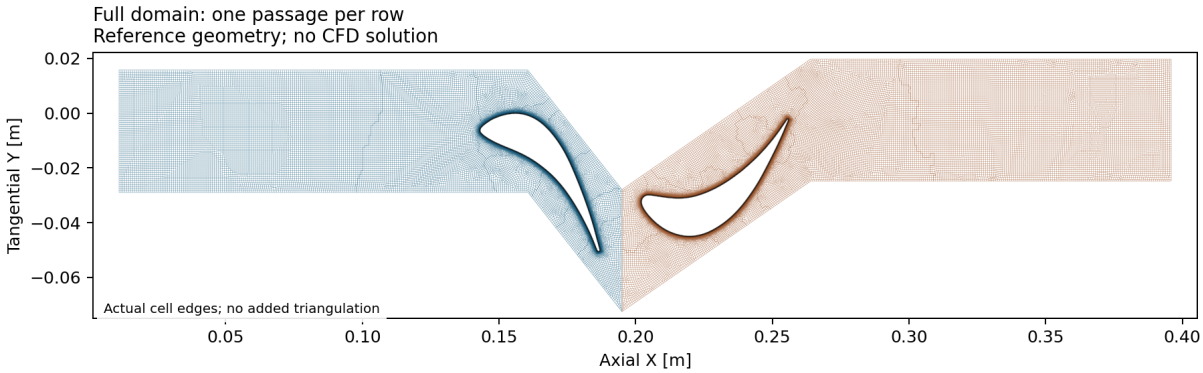
Figure 2: Native leading-edge wall-layer detail for both rows. Coordinates are relative to the actual blade leading edge; no artificial triangular edges are added.



(a) M1, 54287 active cells

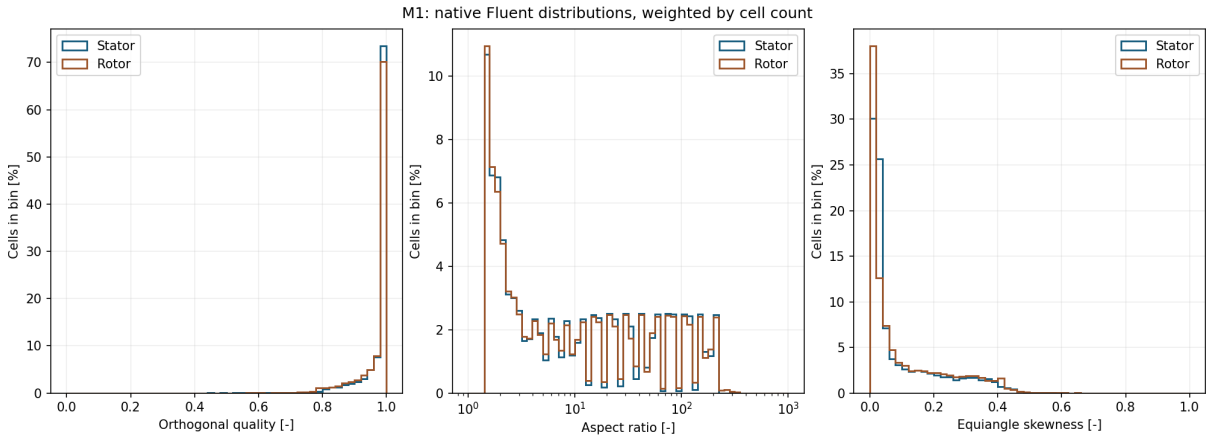


(b) M2, 104831 active cells

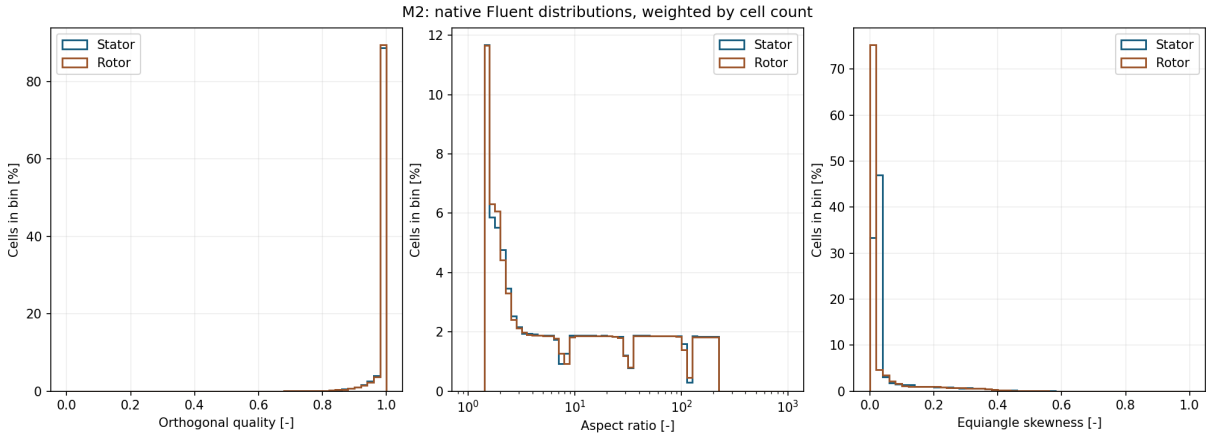


(c) M3, 196569 active cells

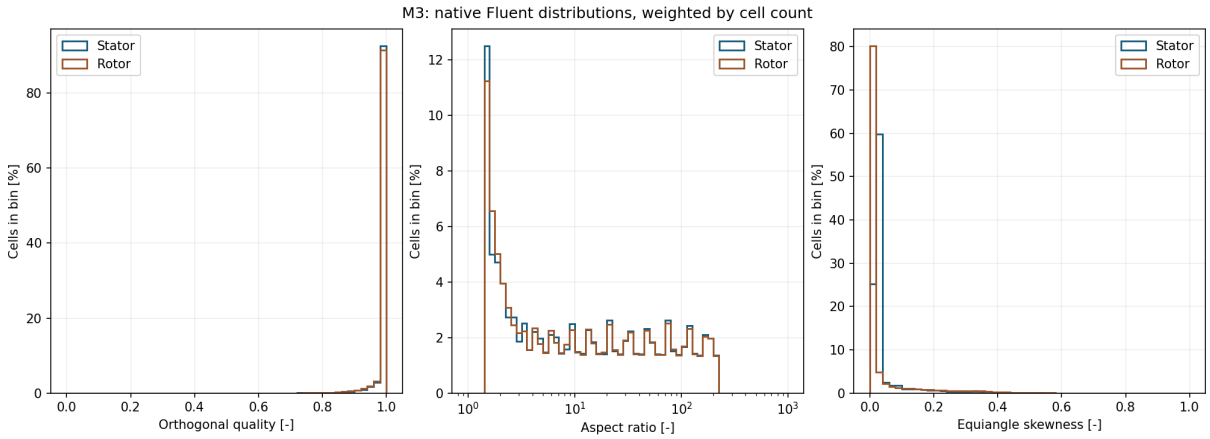
Figure 3: Systematic family from the same geometry. These topology views were recorded at the inherited baseline phase during setup verification; original-position restoration for the accepted Hybrid runs is documented separately.



(a) M1



(b) M2



(c) M3

Figure 4: Orthogonal quality, aspect ratio and skewness distributions. Each histogram is weighted by active cell count, not by cell volume.

5 Baseline integrity, initialization and native restart

5.1 Historical reference and dependencies

The candidate final case/data pair was reopened together in an independent Fluent session. The verified native state is step 3551, physical time 0.010653 s and historical timestep 3e-6 s. Recomputed monitor values agree with the final historical samples. The baseline files remain unchanged.

`inputs/baseline/historical_baseline.cas.h5`: 9897196 bytes. SHA256: `c9f78c24b1a5263f655136f0c1a632ba4276c2eec23a6e1b02622d7276378f7b`

`inputs/baseline/historical_baseline.dat.h5`: 73732083 bytes. SHA256: `1038a1e420ebfa1a47a87d002627b207c05b8016d12ff650990773d737710be1`

The local baseline inputs retain the original file contents under descriptive English names; the checksums above identify the historical pair independently of its filenames. The original Workbench .wbpj and its associated _files directory are preserved together, without internal changes, in the private technical archive outside the repository. A separate portable Workbench archive includes the external STEP, was restored and reopened, and has ten checked internal Location references. The original absolute STEP reference was not changed.

The Fluent project index contained 132 absent historical references, corresponding to 66 case/data checkpoint pairs. The verified final restart did not require those checkpoints. Missing historical files were not recreated using data from different solution times. Native transcripts and SHA256 manifests were retained as provenance records.

5.2 Common checkpoint and automated smoke test

All baseline-mesh tests begin from the same verified checkpoint at t=0.010653 s and step 3551, without resetting time or reinitialising the solution. Output paths are redirected to independent run directories. The six-step smoke test advances dt=3e-6 s from 0.010653 to 0.010671 s, verifies approximately -0.0027 m of rotor translation, unchanged active cells, automatic mesh adaptation disabled, monitor output, coherent saves, rereading and field rendering.

T400 resumed from a checkpoint after 120 steps at t=0.0107422751355 s, global step 3671. Five recomputed reports, including domain mass, matched exactly before further advance. Blade coordinates agreed with the checkpoint HDF5 within approximately 1.49e-8 m. Density, energy, turbulence variables, pressure, temperature and velocity retained their native first and second preceding time levels for second-order continuation. These time-history levels are distinct from the mesh identifiers M1 and M2. The restart occurred entirely within excluded settling.

Restart monitor segments have separate files and explicit time ranges. Coincident samples are merged only if both values and global steps agree. Fluent may add a restart-step suffix to a monitor filename; files are resolved through a unique native header and their actual paths/hashes are recorded. Residual logs were checked step by step to confirm the actual inner-iteration count, independently of whether an iteration-zero row was printed.

5.3 Original-position Hybrid initialization

Preliminary interpolation and inherited-phase initialization trials were excluded from the systematic-mesh comparison. For the accepted campaign, the original rotor geometry was restored before Hybrid Initialization. This provided a consistent initial geometric phase on all three meshes.

For each active mesh, the rotor was restored to the actual geometry of the historical initial t=0 case before setting time and step to zero and applying Fluent Hybrid Initialization. The M3

restoration included a measured Y correction of -0.000900000333786 m. No historical solution data or interpolation field was loaded for initialization. A separate six-step smoke test verified positive fields, motion and restart; the main calculation reread the fresh initialized pair. All systematic-mesh runs used 24 MPI ranks, dt=T/400=7.439594625e-7 s, 30 actual inner iterations per step and disabled automatic adaptation. Restarts preserve time, rotor geometry and BDF2 history.

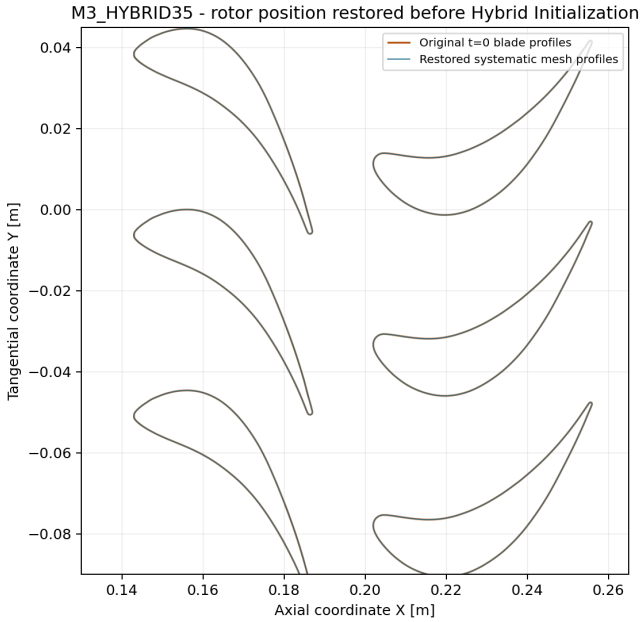


Figure 5: M3 rotor restoration against the actual historical t=0 geometry before Hybrid Initialization. Resetting time without this geometry check would define a different initial position.

6 Numerical verification

6.1 Definitions, statistical windows and recurrence criteria

Three complete blade-passing cycles are compared at the same rotor phase. Cycle means use trapezoidal integration of the native full-precision clock; cycle peak-to-peak amplitudes are calculated before averaging. Phase profiles use 401 points per cycle. The primary quantities are

$$\overline{q} = \frac{1}{3T}\int_{t_a}^{t_a+3T} q(t)\,\mathrm{d}t, \qquad \overline{A} = \frac{1}{3}\sum_{k=1}^{3}\left(\max_{t\in I_k} q(t) - \min_{t\in I_k} q(t)\right). \tag{1}$$

Table 8: Operational diagnostic limits. They are not experimental-accuracy or grid-independence tolerances.

Diagnostic	Operational limit	Meaning
Consecutive cycle means	0.5%	Relative to the later cycle mean
Consecutive amplitudes	5%	Relative to the later cycle amplitude
Phase-profile RMS	10% of amplitude	Cycle-to-cycle recurrence
Mean inlet/outlet imbalance	0.5%	Boundary-flux check
Integrated defect with storage	0.01%	Compressible mass conservation
Phase-matched nodal field RMS	0.5%	Operational field recurrence; not volume weighted

Table 9: Accepted original-position Hybrid campaign at closure. All runs use T/400, 30 inner iterations and 24 MPI processes.

Mesh	Active cells	Periods	Statistics cycles	Monitor recurrence
M1	54,287	35	33-35	Passed recorded limits
M2	104,831	40	38-40	Inlet amplitude: 7.7537% > 5%
M3	196,569	35	33-35	Force amplitude: 7.0685% > 5%

M3 reached 14000 accepted steps and t=0.010415432475 s. Its force amplitudes in cycles 33, 34 and 35 are 199.4099756, 186.2452157 and 185.6334933 N. The last change is only 0.329532%, but the previous change is 7.068509%; the required three-cycle check therefore fails. M2 and M3 remain unconfirmed under the original limits. Only monitor segments bounded by accepted checkpoints are included in these histories.

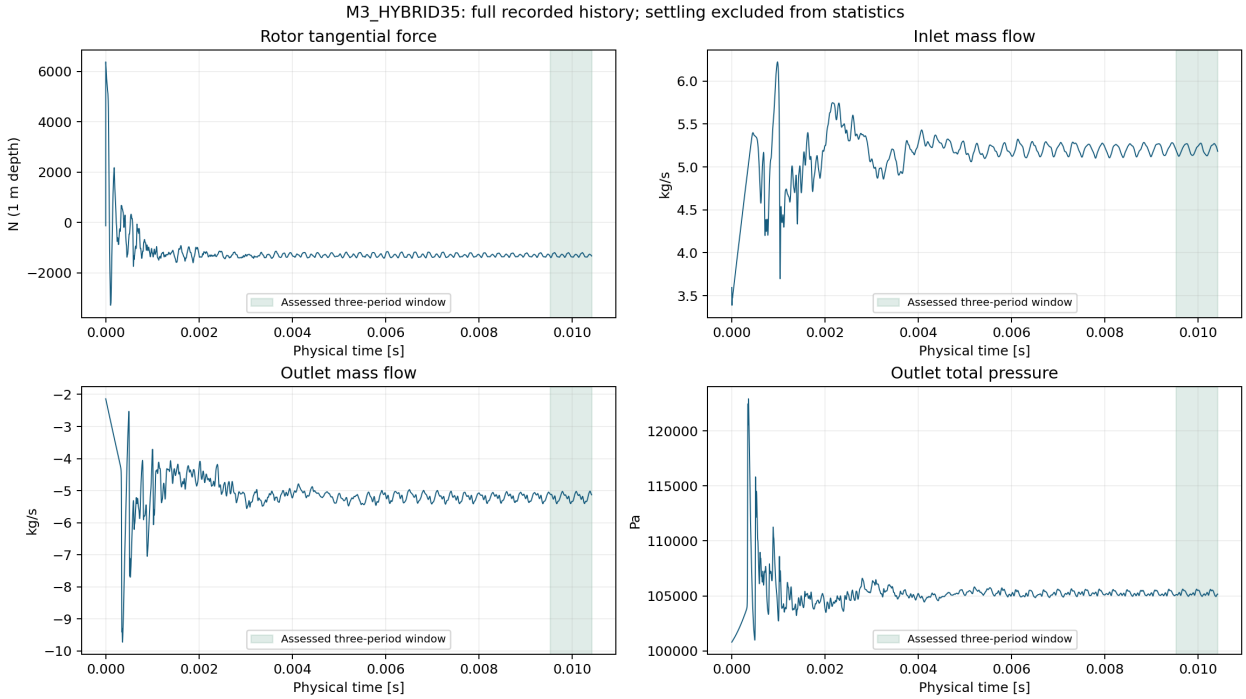


Figure 6: M3 fresh-Hybrid monitor history to 35 periods. The shaded interval is the assessed three-cycle window; the initial transient is not part of its statistics.

6.2 Compressible conservation and native field recurrence

The correct signed mass balance is dM/dt = mass flow at inlet + mass flow at outlet. Integrating the fluxes over a cycle must reproduce the change of mass stored in the domain. A nonzero instantaneous inlet/outlet difference can be physical storage; conservation and periodic recurrence must be checked separately.

The reported integrated mass defect is normalised by the mass transported through the inlet over the same cycle:

$$\epsilon_M = 100 \frac{\left| \int_{t_a}^{t_b} (\dot{m}_{\mathrm{in}} + \dot{m}_{\mathrm{out}})\, \mathrm{d}t - [M(t_b) - M(t_a)] \right|}{\int_{t_a}^{t_b} |\dot{m}_{\mathrm{in}}|\, \mathrm{d}t}. \tag{2}$$

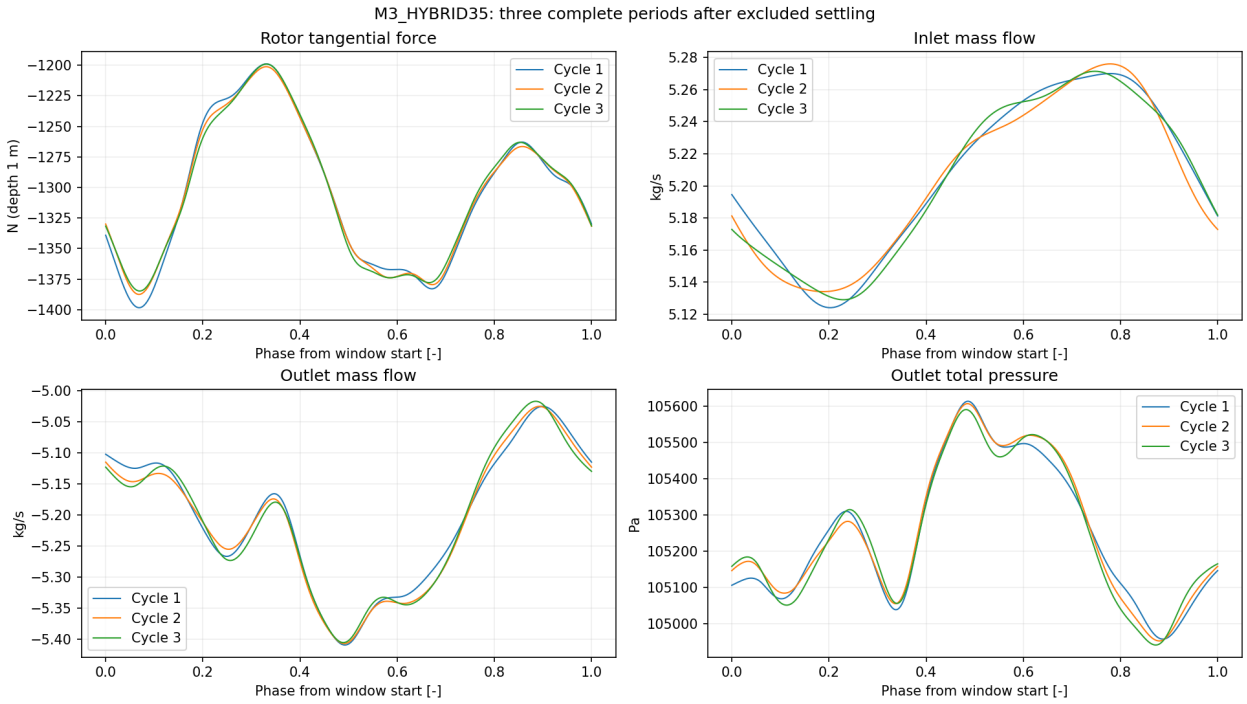


Figure 7: M3 cycle recurrence at common rotor phase. Means are stable, while the force-amplitude test still fails for cycles 33 to 34.

Trapezoidal integration is used for this cycle-integrated diagnostic. The instantaneous BDF2 defect is evaluated separately using the native constant timestep.

Table 10: Recorded conservation, field-recurrence and wall-resolution diagnostics. Passing these checks does not replace the failed monitor-amplitude test.

Mesh	Max integrated mass defect [%]	Max nodal recurrence RMS [%]	Stator / rotor sampled max y+
M1	1.02818e-05	0.133355	0.5837 / 0.5240
M2	1.1913e-05	0.325564	0.4625 / 0.3745
M3	1.55443e-05	0.237437	0.3082 / 0.2682

For matched unique export coordinates, the relative nodal recurrence norm compares an earlier field q_A with the field q_B one period later:

$$\epsilon_q = 100 \frac{\sqrt{N^{-1} \sum_{i=1}^{N} (q_{B,i} - q_{A,i})^2}}{\sqrt{N^{-1} \sum_{i=1}^{N} q_{B,i}^2}}. \tag{3}$$

The denominator is the RMS of the later absolute field, without subtracting its mean; the norm is evaluated separately for each field and fluid zone. Coincident export coordinates can carry differing nodal values. In those cases the assessment uses the largest possible difference at each coordinate and a lower bound for the reference-field RMS to obtain a conservative upper bound over every value selection. No values are averaged or excluded, and the 0.5% limit is unchanged. Identical duplicate values reproduce the unique-coordinate norm above. This exported-node norm is not volume weighted and is not a spatial-discretisation error estimate.

6.3 Inner-iteration sensitivity on the baseline mesh

ITER30 and ITER60 use identical initial fields, mesh, timestep and phase-aligned statistical windows. Residual-based early stopping is disabled to perform exactly 30 or 60 inner iterations

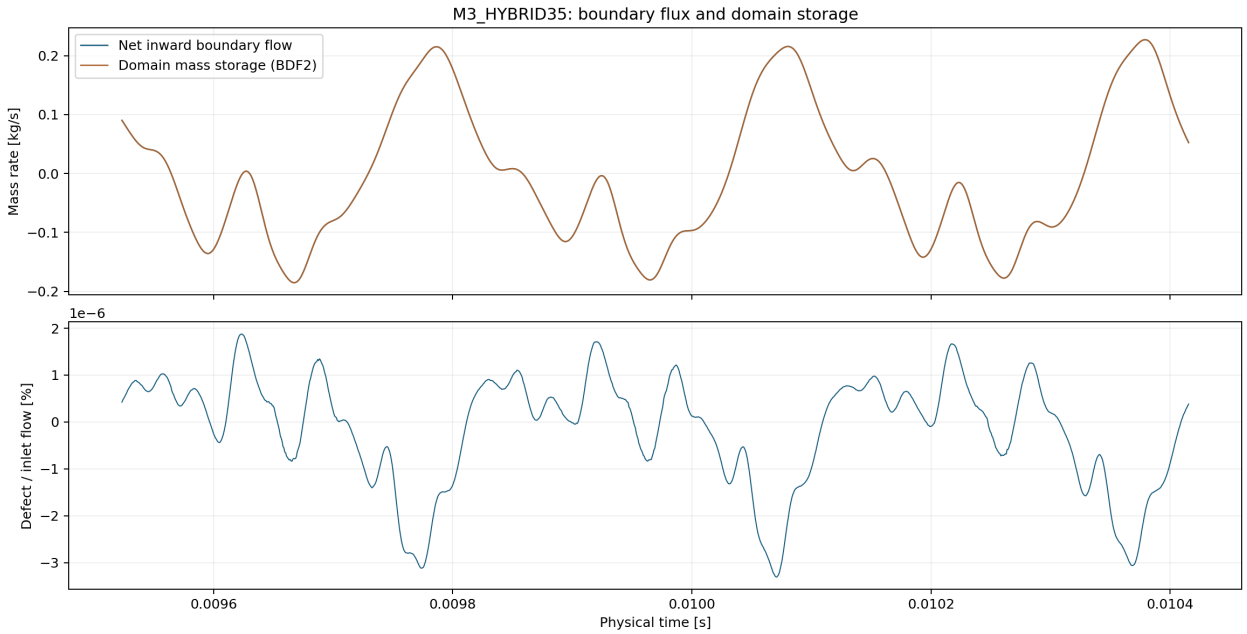


Figure 8: Native boundary flux and BDF2 domain-mass storage over the final M3 window. The small difference is the conservative defect, not the full boundary-flux imbalance.

per step. ITER30 initially used 8 processes for its first 240 settling steps; it then resumed at 24 processes. Its entire statistical window, ITER60 and all subsequent tests use 24 MPI ranks. Rank count is verified; physical-core affinity is not claimed.

Table 11: ITER30/ITER60 comparison on identical initial fields and phase windows

Comparison	**Monitor**	**Mean diff.** [%]	**Amplitude diff.** [%]	**Phase RMS / amp.** [%]
ITER30/ITER60	Rotor tangential force	0.008112165	1.100442	0.6270705
ITER30/ITER60	Inlet mass flow	0.000778778	1.402773	0.6348632
ITER30/ITER60	Outlet mass flow	0.003901606	0.669901	0.4889742
ITER30/ITER60	Outlet mass-flow-weighted total pressure	0.0002717684	0.6225031	0.5274144

The largest mean difference is approximately 0.00812%, amplitude difference 1.403%, and phase RMS 0.635% of amplitude. These satisfy the proposed 0.1% / 2% / 2% operational comparison limits, supporting an initial choice of 30 inner iterations. They remain uncertainty scales when interpreting smaller timestep or mesh differences. Nodal same-phase field norms are not volume-weighted error estimates.

6.4 Temporal sensitivity on the baseline mesh

T100/T200/T400 use T/100, T/200 and T/400, respectively, and end at t=0.012438502710 s at the same rotor phase. Three periods after the timestep change are excluded, followed by three complete statistical periods. Mesh topology remains fixed; rotor translation continues. A comparison with the historical baseline six periods earlier includes field evolution and timestep change and is not a restart-reading error.

T200/T400 means differ by less than approximately 0.0191%, but outlet total-pressure amplitude differs by about 6.559%, and outlet mass-flow amplitude by about 2.724%. The full 0.1% / 2%

Table 12: Timestep-study settings and complete-cycle statistical windows

Run	Timestep [s]	Steps	Excluded periods	Statistics [s]
T100	2.975838e-06	600	3	0.011545751355 - 0.012438502710
T200	1.487919e-06	1200	3	0.011545751355 - 0.012438502710
T400	7.439595e-07	2400	3	0.011545751355 - 0.012438502710

Table 13: Temporal sensitivity of means, amplitudes and phase profiles

Comparison	Monitor	Mean diff. [%]	Amplitude diff. [%]	Phase RMS / amp. [%]
T100/T200	Rotor tangential force	0.0001155694	4.344934	3.891441
T100/T200	Inlet mass flow	0.01569786	0.8079171	2.199374
T100/T200	Outlet mass flow	0.01980105	2.241832	6.330795
T100/T200	Outlet mass-flow-weighted total pressure	0.001658698	6.813504	8.75413
T200/T400	Rotor tangential force	0.01145885	1.89734	1.706666
T200/T400	Inlet mass flow	0.01902148	0.3503656	1.04971
T200/T400	Outlet mass flow	0.01534398	2.723755	2.332154
T200/T400	Outlet mass-flow-weighted total pressure	0.001789769	6.558737	3.666045

/ 2% operational comparison limits are therefore not met. T/400 is retained as the finest tested operational candidate for the spatial study; timestep independence and an asymptotic temporal error are not established. Differences must also be compared with cycle variability and inner-iteration sensitivity.

6.5 Three-mesh sensitivity verified from accepted native monitors

Three-mesh sensitivity was evaluated directly from the accepted native monitor segments. Source hashes, complete step sequences, physical clocks, three-cycle means and cycle amplitudes were checked against the saved analyses. All meshes used T/400 and 30 inner iterations. M1 and M3 were assessed over cycles 33-35, and M2 over cycles 38-40. The windows have equal duration and matching rotor phase, although residual transient differences can remain. The resulting differences therefore describe finite-window mesh sensitivity rather than an isolated spatial discretisation error.

$$\delta_{\overline{q}} = 100\frac{|\overline{q}_c - \overline{q}_f|}{|\overline{q}_f|}, \qquad \delta_{\overline{A}} = 100\frac{|\overline{A}_c - \overline{A}_f|}{\overline{A}_f}. \tag{4}$$

All four M2-M3 mean differences are below 0.15%: force 0.1338%, inlet flow 0.0282%, outlet flow 0.1377% and total pressure 0.0439%. This supports the limited statement that these means are weakly sensitive to the last tested refinement. The amplitude differences remain 6.4156%, 7.1206%, 7.5185% and 5.8633%, respectively. Phase-profile RMS differences reach 8.0058% of the finer-grid amplitude. The oscillating part of the solution is therefore still mesh sensitive.

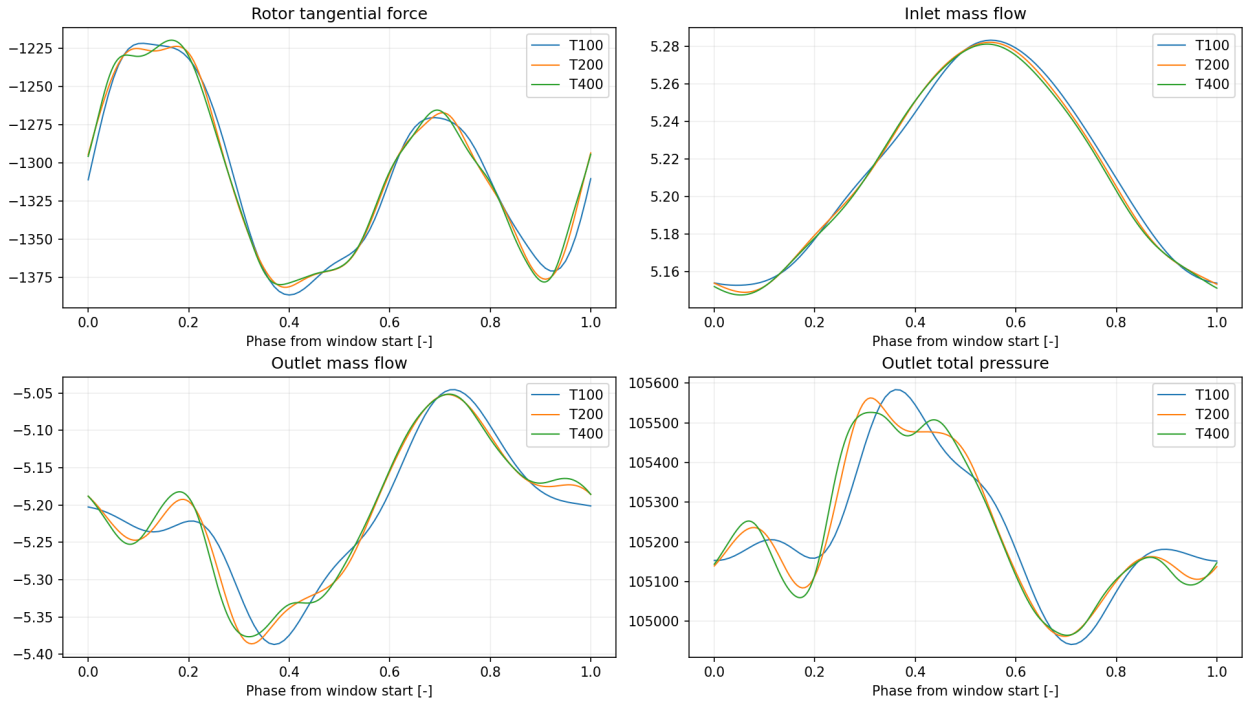


Figure 9: T100, T200 and T400 monitor profiles on the same fixed baseline mesh. Remaining amplitude differences preclude a general timestep-independence claim.

Table 14: Grid sensitivity computed from native monitor histories. Relative differences use the finer-mesh output as denominator; the phase RMS denominator is its mean cycle peak-to-peak amplitude.

Quantity	**M1-M2 mean** [%]	**M2-M3 mean** [%]	**M2-M3 amplitude** [%]	**M2-M3 phase RMS / amplitude** [%]
Rotor tangential force	0.1861	0.1338	6.4156	3.5075
Inlet mass flow	0.1192	0.0282	7.1206	5.6922
Outlet mass flow	0.2176	0.1377	7.5185	4.0220
Outlet total pressure	0.0027	0.0439	5.8633	8.0058

6.6 Asymptotic range and GCI assessment

All four cycle-mean sequences reverse direction between M1-M2 and M2-M3. Total-pressure mean changes increase from 0.0027% to 0.0439% rather than contracting. Force and outlet-flow amplitudes increase monotonically, but their changes do not establish a common asymptotic error model. Inlet and total-pressure amplitudes are nonmonotonic. A small last-pair mean difference alone does not determine the continuum limit or a general error band.

Indicative two-dimensional cell-count refinement ratios are 1.38962 for M1-M2 and 1.36934 for M2-M3, using h proportional to the inverse square root of cell count. They are close to the nominal refinement, but do not prove a single h scale for this anisotropic inflation family. Together with unresolved cycle and timestep sensitivity, the present data do not justify assigning an observed convergence order or GCI. NASA guidance treats assessment of the asymptotic range and the separation of error sources as part of solution verification; the applicable sources are listed in the references.

The computed differences quantify sensitivity to refinement. General grid independence was not demonstrated. Mean outputs were weakly sensitive to the M2-M3 refinement within the assessed

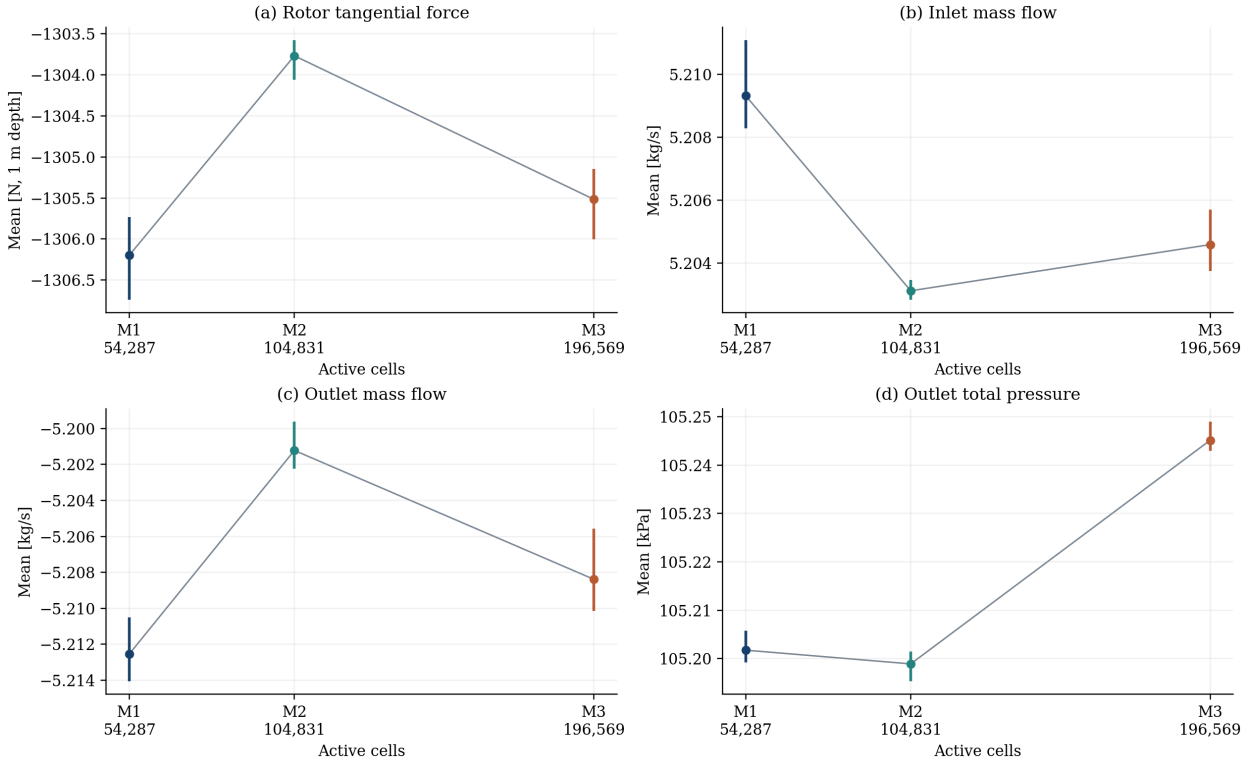


Figure 10: Systematic-mesh cycle means recalculated from native monitors. Vertical ranges span the three observed cycle means; they are not confidence intervals or grid-error bars. Small vertical differences should not be confused with a monotonic asymptotic trend.

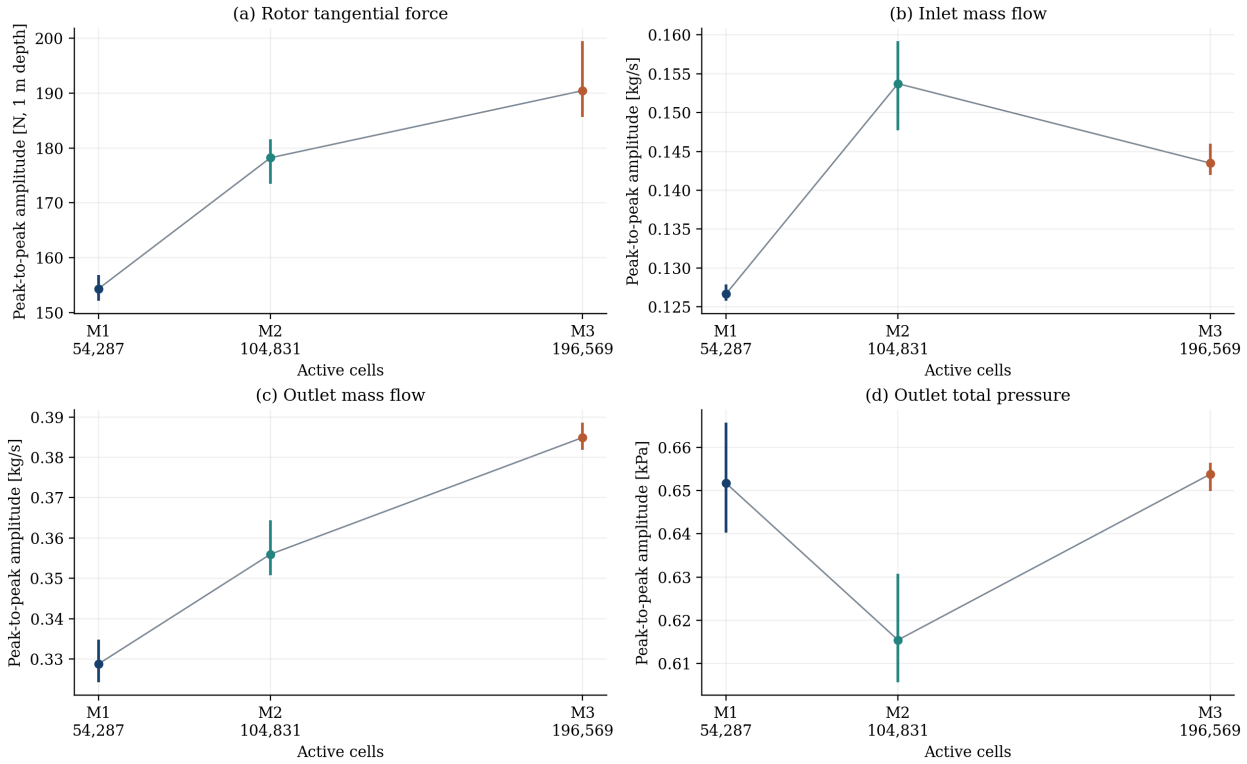


Figure 11: Systematic-mesh peak-to-peak amplitudes. Points are the average of three cycle amplitudes; vertical ranges show the observed cycle variation. Force and outlet-flow amplitudes continue to increase with refinement.

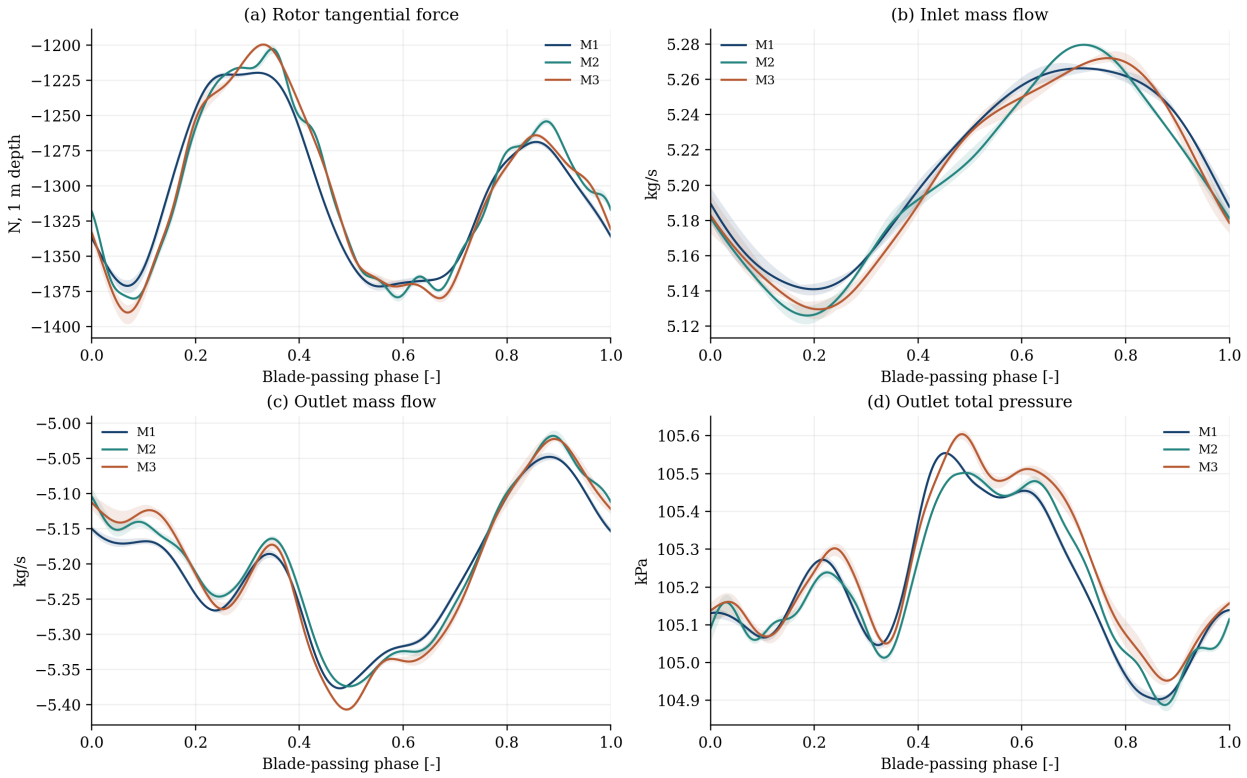


Figure 12: Three-mesh profiles at common blade-passing phase. Lines are the three-cycle phase means; shaded bands span the observed cycles. M2 was sampled five periods later than M1/M3, so residual settling can contaminate mesh differences.

windows, whereas amplitudes and phase profiles retained appreciable differences. No observed convergence order or GCI was assigned.

7 Blade pressure, wall resolution and flow interpretation

7.1 Native wall-face data across the systematic meshes

Final-cycle wall averages use five common phase samples, including the closing endpoint, at intervals T/4 on each mesh. Native wall-face values are plotted directly against axial position. Pressure and suction sides are not joined by an unverified interpolation, and no pointwise wall-error norm is claimed. The pressure distributions almost overlap at the plotted scale; refinement visibly reduces y+ while all sampled maxima remain below one.

For M3, sampled maximum y+ is 0.30818 on the stator and 0.26818 on the rotor. The three-cycle mean tangential force is -1305.516 N, mean inlet mass flow 5.204588 kg/s, outlet mass flow -5.208388 kg/s and mass-flow-weighted outlet total pressure 105245.147 Pa. These remain diagnostic finite-window values, not certified periodic turbine performance.

7.2 Reference-frame and performance limits

Native Mach is absolute and was checked against velocity magnitude divided by sound speed. Blade-wall velocity confirms that rotor mesh motion is applied once. Static pressure is absolute because operating pressure is zero. Total-pressure change across the moving row includes work transfer; it is not presented as a standalone entropy or loss measure. No experimental validation or model-form uncertainty estimate is available.

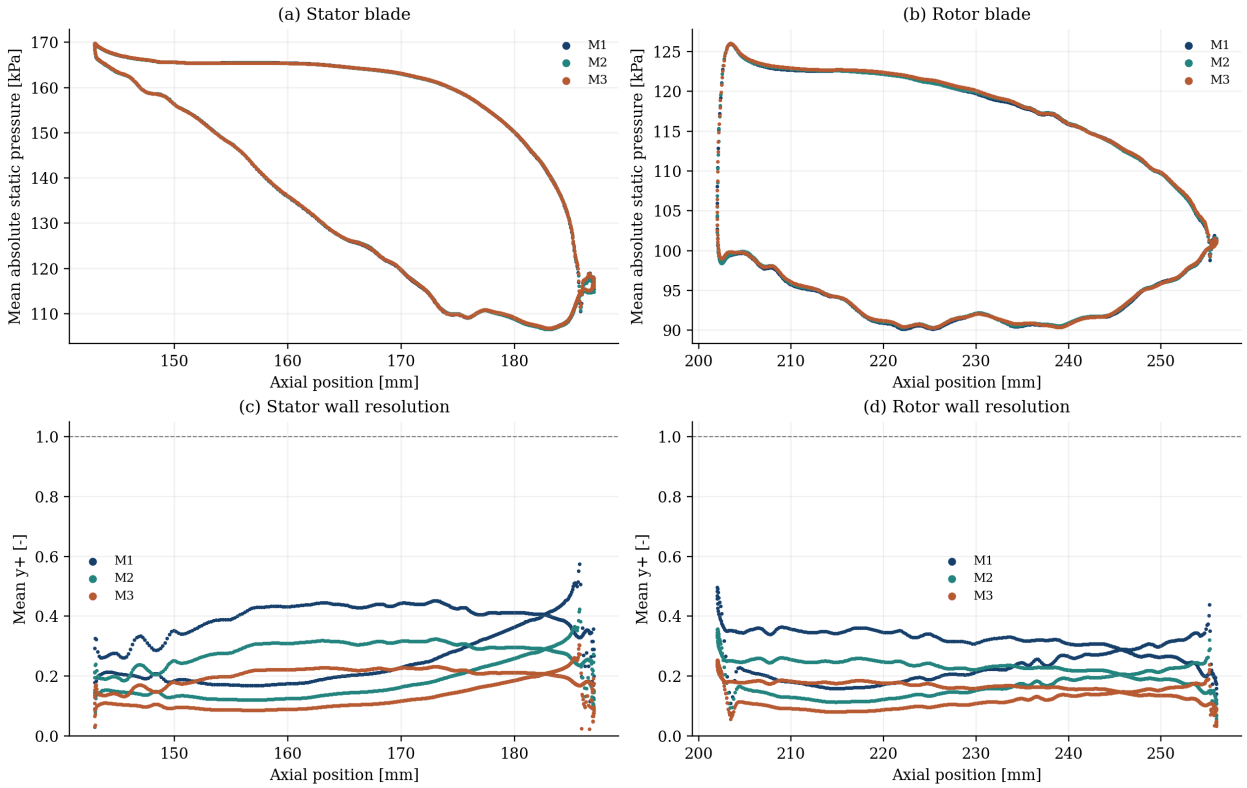


Figure 13: Cycle-mean absolute static pressure and y+ on both blades for M1/M2/M3. Five phase samples per final cycle are used. These profiles show finite-window sensitivity; sampled y+ does not bound unsaved phases.

8 Visualisation and animation products

The selected animation is the fine-mesh M3 absolute-Mach movie, covering the accepted physical interval from 5T to 35T. Mach-only acquisition was used for this transient. The supplied video belongs to the systematic mesh campaign; no pressure movie from that campaign is included.

Table 15: Verified M3 animation configuration and available interval

M3 animation parameter	Verified value
Physical interval [s]	0.001487918925 - 0.010415432475
Native steps / source PNGs	2000-14000, every 10 steps; 1201 PNGs
Physical acquisition interval	7.439594625e-6 s; 40 frame intervals per pitch period
Resolution / playback	1920 x 1080; 25 FPS; 48.04 s
Mach legend	Fixed 0-1.2; values above 1.2 saturate the top colour and this is disclosed
Camera and repeat geometry	Fixed three-pitch crop; verified native periodic instances on both rows
Time annotation	Actual Fluent Solution time, 12 decimal places
Encoder	Offline H.264 from unchanged native Fluent PNGs; no CFD and no Fluent Playback

Only frames associated with the accepted main solution were included in the M3 movie. The complete sequence from t=0 was not verified and is not reported as an available result. The verified interval is 5T-35T. The main 30T-35T continuation was checked independently against native time, mesh geometry, solver settings, report values and BDF2 histories before advancing.

The final MP4 was decoded for beginning, quarter-phase motion, middle and end samples. Blade

rows, fixed legend size, camera, physical clock and annotation were visually checked. Image hashes and dimensions were checked for all 1201 source frames. Sampling inspection does not establish temporal smoothness at every frame or periodic convergence. Camera cropping naturally shows partial blades at its upper and lower boundaries as the row translates. Playback FPS is distinct from physical acquisition rate.

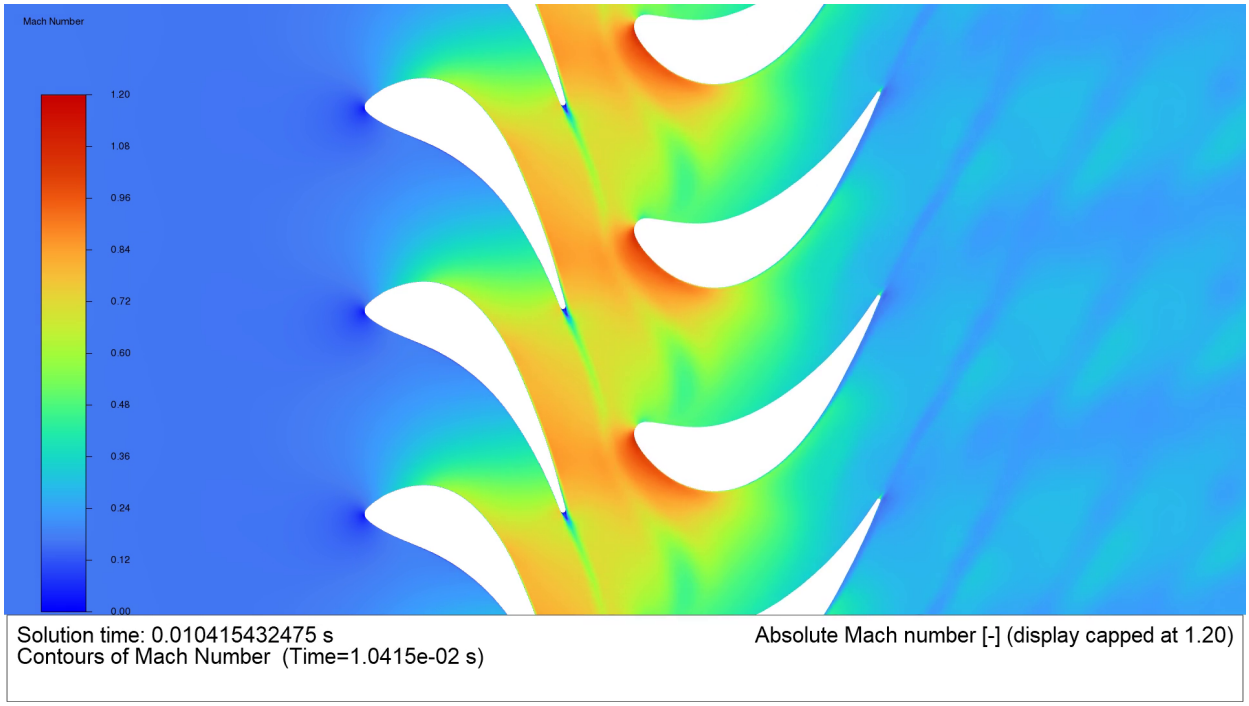
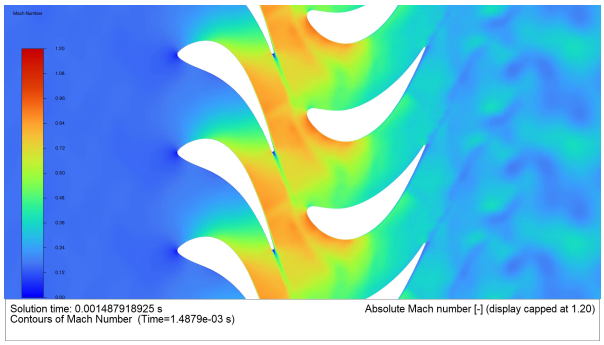
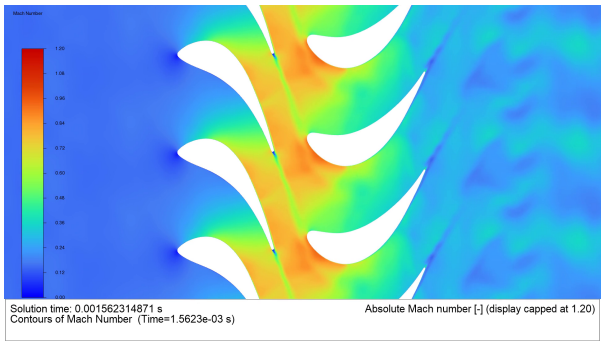


Figure 14: Fine-mesh absolute Mach field at 35T, with three visible pitches and a fixed 0-1.2 display cap. Upper-colour saturation is explicitly disclosed. The annotation uses the actual physical Fluent clock.



(a) M3 at 5T



(b) M3 at 5.25T

Figure 15: Quarter-period rotor displacement in decoded native-PNG movie frames. Camera, periodic crop and colour-bar geometry are unchanged. This visual check is distinct from the unresolved periodicity of the flow.

9 Discussion and limits of the present results

Residual amplitude evolution can remain after cycle means and mass conservation appear stable. The available data do not uniquely separate transient evolution, sampling sensitivity and numerical error. Mesh quality and sampled wall resolution provide useful numerical checks, but do not establish spatial independence or physical accuracy.

Table 16: Final interpretation by verification topic

Topic	Final conclusion
Historical baseline	Coherent saved reference; fixed topology and automatic adaptation disabled
Restart and automation	Verified smoke tests and strict native continuation; historical failures retained separately
Iteration sensitivity	30/60 comparison supports 30 on the baseline mesh; not a separate proof for M3
Timestep sensitivity	T200/T400 amplitude criterion not met; timestep independence unconfirmed
M1 original-position Hybrid	35 periods; recorded operational settling checks passed
M2 original-position Hybrid	40 periods; inlet-amplitude recurrence still fails the three-cycle limit
M3 original-position Hybrid	35 periods; force-amplitude recurrence still fails one cycle pair
Spatial convergence / GCI	Not established; no observed order or error bar assigned
M3 video	Verified available 5T-35T transient interval; complete t=0 sequence unverified
Validation	No experimental validation and no certified turbine performance claim
Simulated duration	M1: 35 periods; M2: 40 periods; M3: 35 periods; finite-window limitations retained

The four mean quantities differ by at most 0.138% between M2 and M3, while amplitude changes remain above 5.8%. These observations indicate different sensitivities for different quantities of interest. The nonmonotonic three-mesh means do not support a formal convergence-order estimate or a GCI error bar.

10 Conclusions

The historical baseline and Workbench dependencies are preserved. Native restarts and original-position Hybrid starts are traceable. All systematic meshes have positive cell volumes, with recorded quality metrics and sampled wall-resolution data. The workflow produced accepted monitor histories, conservation checks, figures and a Mach movie covering the documented interval.

The three-grid assessment was based on saved native data. Last-pair means were weakly sensitive to refinement, while amplitudes and phase waveforms retained appreciable mesh sensitivity. Periodic settling on M2/M3 and temporal independence remained unresolved. The reported durations were 35 periods for M1, 40 for M2 and 35 for M3; general spatial independence and an asymptotic GCI were not established.

A Data availability and reproducibility

The numerical dataset comprises the historical reference pair, the systematic mesh files, coherent final case/data pairs, accepted native monitor segments, sampled phase fields, wall-face exports and numerical verification tables. Solver settings, timestep, internal iteration count, mesh size, reference depth and physical-time intervals are recorded alongside each result. Intermediate checkpoints and superseded development outputs are retained separately in the private technical archive.

The repository separates flat Python modules in `workflow/`, configuration in `config/`, geometry and source attribution in `geometry/`, and selected numerical tables, figures and animations in `results/`. Only the final PDF report is distributed in `docs/report/`; document preparation is separate from the numerical workflow. Large native input files in `inputs/` and solver output directories in `runs/` are local resources excluded from normal Git history. Run identifiers and SHA256 manifests associate the publication figures and tables with their numerical sources. ANSYS software and the Python runtime are not redistributed.

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